

RESEARCH METHODOLOGY

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ANALYSIS OF HOAX NEWS SENTIMENT USING NAIVE BAYES

Who Needs that Idea?

The research idea "Analysis of Hoax News Sentiment using Naive Bayes" is needed by:

- **The general public** → to help detect circulating hoax news, so they are not easily influenced.
- **The government/relevant institutions (Kominfo, BSSN, Police)** → as a tool to classify hoax news that can be used in digital literacy and prevent the spread of fake news.
- **Online media and journalists** → to conduct initial filtering of news or public comments.
- **Academics and researchers** → interested in sentiment analysis, NLP (Natural Language Processing), and simple machine learning for Indonesian language case studies.
- **Application developers** → especially those interested in developing desktop, web, or mobile-based hoax news detection applications.

Novelty

- **ANALYSIS OF HOAX NEWS SENTIMENT USING NAIVE BAYES**
 1. Development of a Python desktop application and user-friendliness testing (UAT) with users.
 2. Sentiment classification with three classes (positive, neutral, negative) regarding hoax news.
 3. Focus on "digital evidence" of hoax news (can be more tangible/digital evidence data).
 4. Metrics such as accuracy, precision, and recall are measured numerically (similar to quantitative research).
- **TELKOMNIKA**
 1. Integration of hoax detection and sentiment analysis on hoax news.
 2. Naive Bayes optimization using PSO to improve performance.
 3. Focus on hoax news spread on social media.
 4. Extract "hidden sentiment" in hoax news after detection.

Research Gap

- **ANALYSIS OF HOAX NEWS SENTIMENT USING NAIVE BAYES**
 1. It may not have used optimizations like PSO or additional feature weighting techniques, making it less effective than optimization research.
 2. The number of UAT respondents was very small (3 people according to the summary) → a weakness in real-user validity.
 3. There was no comparison with other algorithms like PSO-NB, SVM, RF, or deep learning to determine which performed better.
 4. The domain of "digital evidence of hoax news" may be unclear: does it include social media, online news, forums, WhatsApp, etc.? The diversity of sources can create gaps.
 5. Dataset limitations: how much hoax news data was used, the representativeness, and the verification of hoax/sentiment labels.
- **TELKOMNIKA**
 1. Small dataset (only ~30 hoax news stories), so generalizability is limited.
 2. No user usability/ease-of-use evaluation of the application; more focus on algorithms and performance.
 3. PSO optimization, but perhaps less discussion of the desktop/web application implementation side.
 4. Hoax news domains are specific to social media, possibly lacking domain diversity (politics/health/job openings, etc.).

Jurnal Link (Sinta 1)

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